

Regularization in Machine Learning

Lasso Regression

- A regression model which uses the L1 Regularization technique is called LASSO (Least Absolute Shrinkage and Selection Operator) regression.
- Lasso Regression adds the “absolute value of magnitude” of the coefficient as a penalty term to the loss function(L).
- Lasso regression also helps us achieve feature selection by penalizing the weights to approximately equal to zero if that feature does not serve any purpose in the model.

Regularization in Machine Learning

Lasso Regression

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m |w_i|$$

- *where,*
- *m – Number of Features*
- *n – Number of Examples*
- *y_i – Actual Target Value*

Regularization in Machine Learning

Ridge Regression

- A regression model that uses the **L2 regularization** technique is called **Ridge regression**.
- **Ridge regression** adds the “*squared magnitude*” of the coefficient as a penalty term to the loss function(L).

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m w_i^2$$

Regularization in Machine Learning

Elastic Net Regression

- This model is a combination of L1 as well as L2 regularization.
- That implies that we add the absolute norm of the weights as well as the squared measure of the weights.
- With the help of an extra hyperparameter that controls the ratio of the L1 and L2 regularization.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \left((1 - \alpha) \sum_{i=1}^m |w_i| + \alpha \sum_{i=1}^m w_i^2 \right)$$